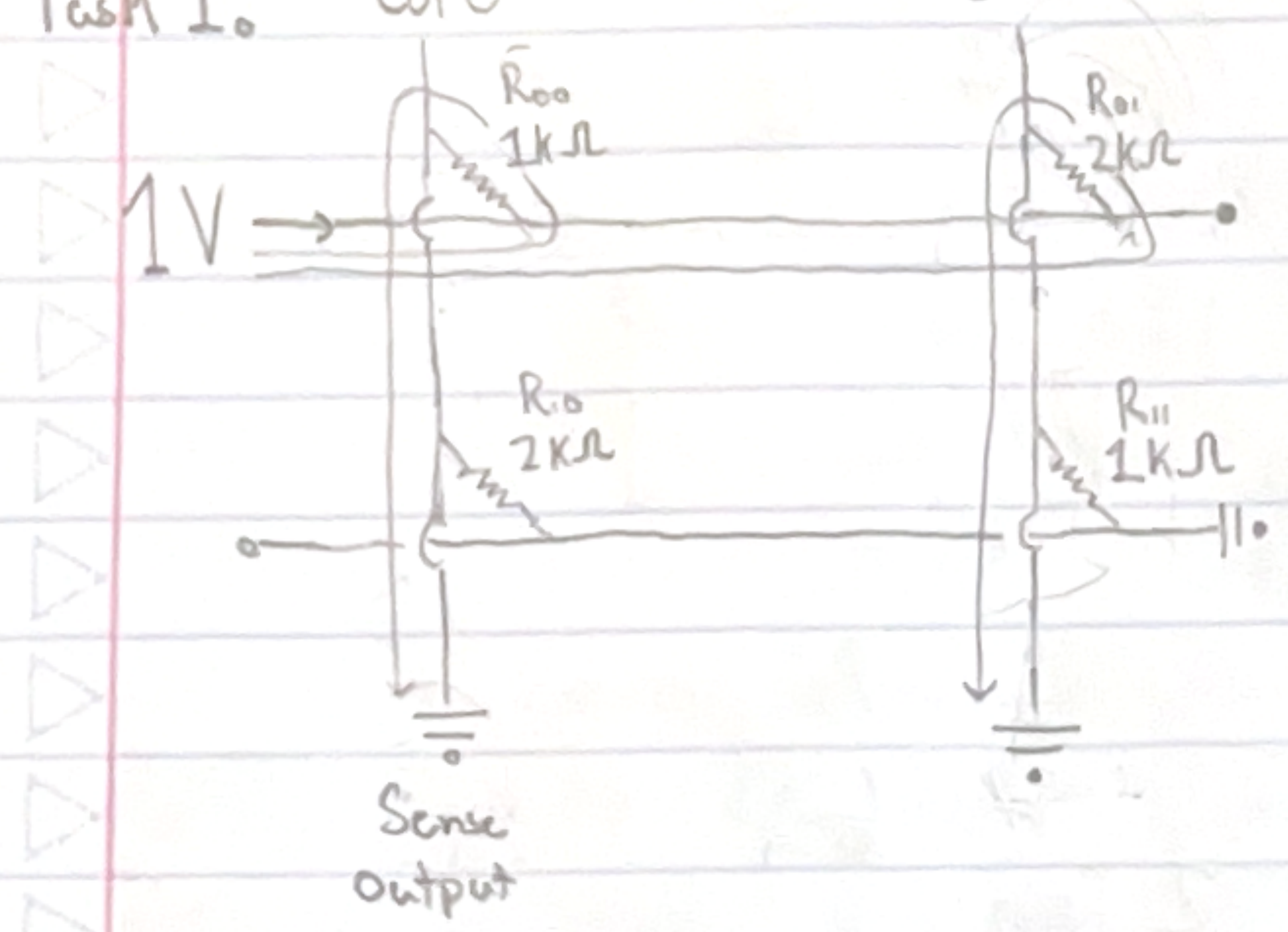
Codefest 6. CMAN

Task 1.



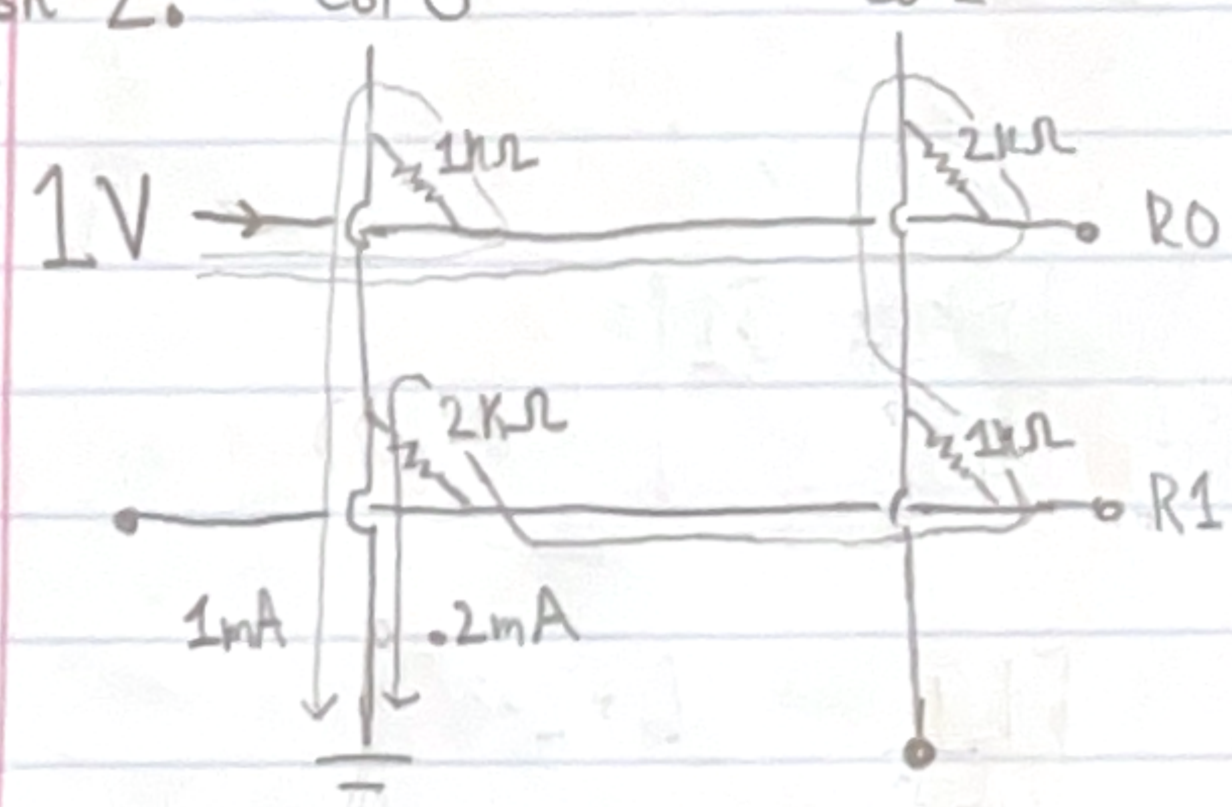
Effective resistance = $1/(\frac{1}{1k\Omega}+\frac{1}{2k\Omega})$

Effective resistance = 666Ω

Total flow = 1.5mA

Since resistance ratio is 1 to 2, 2/3 flows through R_{00} and 1/3 flows through R_{01}. $I_0 = 1mA$, $I_1 = .5mA$.

Task 2.



Effective resistance = $1/(\frac{1}{1k\Omega}+\frac{1}{2k\Omega+1k\Omega+2k\Omega})$ = 833Ω.

Total flow = 1V/833Ω = 1.2mA

with sneak current, 20% more flow. Sneak path flow is .2mA, so

V(col1) = 1V − 2kΩ · .2mA = 1V − .4V = .6V and

V(row1) = .6V − 1kΩ · .2mA = .6V − .2V = .4V.

Task 3. The sneak path provides an additional undesired component to the dot product. This is an issue for large matrices, as the larger the matrix, the more sneak paths will become available. This will result in larger and larger perterbutions to the correct output as matrix size increases.